

Chapter 30 Back Matter Check

Contents

30 Supergraphs	1
Problems	1
References	1

30 Supergraphs

Problems

1. Use Eq. (30.2.18) to calculate the propagators of the spinor and auxiliary components of a chiral superfield.
2. Consider a supersymmetric theory of a single left-chiral superfield Φ , with Lagrangian density

$$\mathcal{L} = \frac{1}{2} [\Phi^* \Phi]_D + 2 \operatorname{Re}(g[\Phi^3]_{\mathcal{F}}),$$

with g an arbitrary complex constant. Use the supergraph formalism to calculate the one-loop contribution to the quantum effective action. Express the answer as an integral over coordinates and over a single Grassman coordinate θ .

3. What is the superpropagator for a gauge superfield $V(x, \theta)$ in a supersymmetric Abelian gauge theory with kinematic term (27.3.17)?

References

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2. M. T. Grisaru, W. Siegel, and M. Roček, *Nucl. Phys.* **B159**, 429 (1979).